Mohamed Shamroukh Mohamed Mahmoud

Doctoral Researcher in Geospatial AI & Urban Planning | Loughborough University, UK.
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Education

PhD in Urban Planning | Loughborough University *Apr 2024 -Present*

Thesis: "Towards Pedestrian-Friendly Cities: Employing Geospatial Artificial Intelligence for Urban Planning"

- Developing a city-wide Geospatial Pedestrian Activity Engine (GPAE) integrating WiFi sensor data, machine learning ensemble methods (LightGBM, XGBoost), and multi-source mobility data for monitoring and evaluation of urban environments.
- Developed a location-aware calibration framework achieving 85.8% accuracy across eight urban sites.
- Conducting systematic evidence reviews, field observations, and secondary data analysis on pedestrian monitoring technologies and urban mobility datasets.
- Supervised by Dr Asya Natapov and Dr Taimaz Larimian, School of Architecture, Building and Civil Engineering.

MA in Geographic Information Science | South Valley University, Egypt *Jan 2022*

Thesis: "Developing a Smart Automated GIS Map for Public Services in Qena City", Grade: Excellent

BA in Geography (Geographic Information Science) | South Valley University, Egypt *Sept 2017*

Grade: 88.91%, Very Good (Honours)

Professional Memberships and Recognition

- Associate Fellow of the Higher Education Academy **(AFHEA)**, Advance HE, 2026.
- Associate Member, Python Software Foundation **(PSF)**

Research Funding & Grants

- Fully funded PhD Scholarship, Loughborough Town Deal project, co-funded by the Department for Levelling Up, Housing and Communities (DLUHC) and Loughborough University. £140,000 (2024-2027)
- Co-Investigator (Analyst), DfT Evidence Reviews and Secondary Data Analysis (CCZZ25A21), with a multidisciplinary team from Loughborough University, funded by the Department for Transport (DfT)- Social and Behavioural Research (SBR) Team. £1,500,000 (2026–2029)

Research & Professional Experience

Teaching Assistant | Loughborough University *Apr 2024 - Present*

- Teaching lab sessions and contributed to module teaching in Urban Planning, GIS, Design Skills, and laser scanning within the School of Architecture, Building and Civil Engineering.
- Supervising undergraduate dissertation students (2025 to 2027) on geospatial and urban planning topics, advising on research skills and design, spatial analysis, data interpretation, and academic writing.

Research Assistant, Clean Energy Projects| De Montfort University (Part-time) *Oct 2026 - Present*

- Supporting research and data analysis for clean energy and sustainability projects.
- Contributing to project development, data management, and research outputs.
- Applying quantitative and computational methods to analyse energy-related challenges.
- Supporting the preparation of reports, publications, and stakeholder engagement activities.

DEBIAS Network Fellow | Liverpool University *Jul 2026 - Present*

- Contributing to the development and standardization of digital trace data protocols and methodologies, addressing data quality, reliability, GDPR compliance, FAIR data principles, and interoperability across datasets.

Net Zero Engagement Research Fellow | De Montfort University (Part-time) *Feb 2026*

- Designed and delivered a Solar Streets Opportunity Assessment for four Leicestershire districts (Blaby, Melton, Northwest, and Oadby & Wigston), evaluating rooftop solar installation potential across thousands of residential and commercial buildings
- Built an automated deep learning pipeline using YOLO and satellite imagery (XYZ tiles) to detect and classify rooftop solar panels at district scale, without the need for proprietary data

Researcher | The Impact Hub, Loughborough University *Apr 2024 - Mar 2026*

- Co-developed the "Loughborough in Motion" interactive geospatial dashboard alongside Dr Asya Natapov and Dr Mohamed Forghani, integrating pedestrian footfall, mobility, lightening, and urban activity data.
- Contributed to a town-scale digital observatory monitoring and evaluating urban dynamics, mobility patterns, and place-based interventions in Loughborough

Research Assistant | South Valley University *Feb 2022 - Mar 2024*

- Conducted spatial data analysis and GIS-based research supporting faculty research programmes.
- Co-authored multiple peer-reviewed journal articles integrating GIS, GeoAI, and geospatial programming.

Teaching Assistant | South Valley University *May 2019 - Jan 2022*

- Delivered lab courses in GIS, Remote Sensing, Map Algebra, SQL, and Land Use Analysis.
- Supported student learning and assessment across undergraduate geography and GIS programmes.

Publications

Full publication list and citation metrics available at: Google Scholar

Journal Articles

- **Shamroukh, M**., Natapov, A., & Larimian, T. (2026**). A predictive geospatial engine for city-wide pedestrian-centric urban regeneration**, Travel Behaviour and Society, 46, 101422. https://doi.org/10.1016/j.tbs.2026.101422
- **Shamroukh, M**., Natapov, A., & Larimian, T. (2026). **Mitigating sensor bias: A location-aware calibration framework for pedestrian analytics**. Comput.Urban Sci. 6, 65 (2026). https://doi.org/10.1007/s43762-026-00295-0
- **Shamroukh, M**., Natapov, A., & Larimian, T. (2025). **Reliability and scalability of pedestrian monitoring practices: A systematic review**. IEEE Access, 13, 147679–147689. doi: 10.1109/ACCESS.2025.3599283
- **Shamroukh, M**., & Aziz, M. A. (2023). **Integrating GIS, GeoAI, and Geospatial Programming to Support Methods of Spatial Data Analysis.** Journal of Qena Faculty of Arts, 32(60), 135-162. doi: 10.21608/qarts.2023.224800.1721.
- **Shamroukh, M**., Aziz, M. A., Mahmoud, I. D., & Othman, H. S. N. (2023). **Towards Developing a Web-Based GIS Using Open-Source Software.** Journal of Qena Faculty of Arts, 32(59), 505-534. doi: 10.21608/qarts.2023.215768.1694.

Book Chapters

- **Shamroukh, M**., Natapov, A., & Larimian, T. (2027). **Mapping pedestrian flows for urban vitality**. In: Shekhar, S., Xiong, H. (eds) Encyclopaedia of GIS. Springer, Boston. (accepted).
- **Shamroukh, M**., Natapov, A., & Larimian, T. (2026). **Geospatial Pedestrian Time-series Intelligence for Smart Eco-city Infrastructure Optimisation and Resilience**. Book chapter submitted to AI Frontiers for Smarter Eco-Cities: Deep Learning, Generative, and Foundation Models for Environmental, Geospatial and Time-Series Intelligence (Ed. S. Bibri), Elsevier. (accepted).

Peer Review

- Peer reviewer for international journals and books in GIScience, urban planning, mobility and smart cities.

Open-Source Contribution

QGIS Pip Manager Plugin (ID: 3767) | plugins.qgis.org/plugins/pip_manager *Mar 2025 - Present*

- Co-developed a QGIS plugin enabling Python package management via a graphical interface, improving accessibility for non-technical users.
- Achieved 8,600+ downloads since release (March 2025).

Conferences & Invited Talks

Conferences & Presentations

- Loughborough Doctoral College Showcase 2026. Mapping Where People Walk: A City-Wide GeoAI Engine for Pedestrian Flow Prediction.
- GeoAI 2026, Ghent, Belgium. GeoAI for city-wide pedestrian flow mapping and urban regeneration.
- CUPUM 2025, London. Spatio-temporal analysis of pedestrian activity in Loughborough using Wi-Fi data.
- AESOP 2025, Istanbul. Data-driven prioritising of pedestrian-oriented infrastructure: A framework for resource allocation.
- EMDOC 2025 PGR Conference. Mapping Loughborough's Pedestrian Activity Using WiFi Data.
- RGS-IBG 2025, Birmingham. From Data to Decisions: Understanding Pedestrian Flow with WiFi Scanner Data.
- Loughborough Doctoral College Showcase 2025. Mapping Loughborough's Pedestrian Activity Using WiFi Data.

Invited Talks and workshops

- Invited to present research on urban analytics and GeoAI to delegates from the Gauteng City-Region Observatory (GCRO), south Africa, at the Institute for Advanced Studies, Loughborough University, November 2026.
- DEBIAS Workshop and Symposium, on the 7th and 8th of July2026 in Liverpool, UK.
- Public engagement event with local residents - TOWN Observatory at Loughborough Heritage Open Weekend (England's largest festival of history and culture), 15 Market Place, Loughborough, UK. September 21, 2025.
- Knowledge Exchange Workshop, Gauteng City-Region Observatory (GCRO) & The Impact Hub, Loughborough University, April 2025.
- Ibn Tofail University, Morocco, December 2021.
- Umm Al-Qura University, Saudi Arabia, November 2021.
- University of Diyala, Iraq, March 2021.
- Egyptian Geographic Society, Egypt, November 2021.

Awards & Honours

- **Doctoral College Poster Award**, " Mapping Where People Walk: A City-Wide GeoAI Engine for Pedestrian Flow Prediction.", Loughborough University (2026); one of four award recipients selected from 74 participants across the University.
- **School of Architecture, Building and Civil Engineering Poster Award**, Loughborough University (2026); one of two award recipients selected from 25 participants across the school.
- **Doctoral College Poster Award**, "Mapping Loughborough's Pedestrian Activity Using WiFi Data", Loughborough University (2025); one of four award recipients selected from 67 participants across the University.
- **School of Architecture, Building and Civil Engineering Poster Award**, Loughborough University (2025); one of two award recipients selected from 20 participants across the school.
- **Open-Source Geospatial Foundation** (OSGeo), "Pitch Your Research" Geo for All Student Competition, December 2021.
- **South Valley University President Award**, Student Competition, 2017.

Technical Skills

Category	Tools & Technologies
Programming & Data Analysis	Python (GeoPandas, Pandas, Scikit-Learn, LightGBM, XGBoost, SHAP, Rasterio, OpenCV, YOLO), SQL, HTML
GIS & Remote Sensing	QGIS, ArcGIS, ArcGIS Online, Google Earth Engine, ERDAS Imagine, Global Mapper
Machine Learning & Ai	Ensemble methods, deep learning, computer vision, cross-validation, predictive modelling
Web & Data Tools	Django, Leaflet, Jupyter Notebook, GitHub/Git, PostgreSQL/PostGIS
Data Visualisation	Matplotlib, Seaborn, Plotly, interactive geospatial dashboards
Secondary Data Analysis	WiFi sensor data, behavioural observation, footfall counters, satellite imagery, multi-source spatial data integration
Transferable Skills	Grant writing, student supervision, public engagement, interdisciplinary collaboration, science communication
Language	English (professional fluency)

References

Dr Asya Natapov | a.natapov@lboro.ac.uk **Dr Taimaz Larimian** | t.larimian@lboro.ac.uk